

Business KPI Assistant

1. Problem Statement

Traditional business data analysis often requires users to manually clean datasets, write formulas, create charts, and interpret numerical results. Non-technical users may find it difficult to extract meaningful insights from large Excel or CSV datasets, while manually preparing reports can be time-consuming and prone to errors.

To overcome these limitations, a **Business KPI Assistant** is proposed. The system is an AI-powered analytics tool that allows businesses to upload sales or operational datasets and interact with their data using natural language. The system automatically analyzes the uploaded data, identifies important Key Performance Indicators (KPIs), generates relevant visualizations, detects trends and anomalies, and provides actionable business insights.

The proposed system aims to simplify data analytics by allowing users to ask questions such as “*Which product had the highest sales?*”, “*Why did revenue decrease last month?*”, or “*Which region is underperforming?*” without requiring them to manually write formulas or analytical queries.

2. Input

The proposed system accepts structured business datasets and natural-language queries from users.

2.1 Sales / Operational Data

The system accepts business data in formats such as:

- CSV
- Excel (.xlsx)

Example data fields may include:

- Product Name
- Product Category
- Sales
- Revenue
- Quantity Sold
- Profit
- Region
- Customer Segment
- Date
- Order ID
- Operational Metrics

Purpose:

- Business performance analysis
- KPI calculation
- Trend analysis
- Comparative analysis
- Visualization generation

2.2 Natural Language Query

Users can interact with the uploaded dataset using natural-language questions.

Examples:

- Which product had the highest sales?

- Which region generated the most revenue?
- Why did revenue drop last month?
- Which products are underperforming?
- What is the monthly sales trend?
- Are there any unusual sales values?

Purpose:

- Understand user intent
- Identify required metrics
- Select relevant dataset columns
- Determine appropriate analytical operation
- Generate meaningful business insights

2.3 Temporal Data

The system uses time-related information available in the dataset, such as:

- Date
- Month
- Quarter
- Year
- Week

Purpose:

- Time-series analysis
- Growth analysis
- Seasonal trend identification
- Period-to-period comparison
- Forecasting

2.4 Historical Business Data

Previously recorded business data can be analyzed to identify historical patterns and changes.

Purpose:

- Historical performance analysis
- Trend identification
- Growth comparison
- Anomaly detection
- Future performance analysis

3. Output

The system generates analytical outputs based on the uploaded dataset and user's query.

3.1 KPI Identification and Calculation

The system automatically identifies and calculates relevant business KPIs such as:

- Total Revenue
- Total Sales
- Total Profit
- Profit Margin
- Average Order Value
- Units Sold
- Revenue Growth
- Sales Growth
- Customer-related metrics

Purpose:

- Provide a quick overview of business performance
- Track important performance indicators
- Support data-driven decision-making

3.2 Natural Language Business Insights

The system converts analytical results into easy-to-understand explanations.

For example:

User: Which region is underperforming?

The system analyzes regional performance and provides an explanation such as:

“The South region recorded the lowest revenue this quarter, with a 14% decline compared with the previous quarter.”

Purpose:

- Simplify complex analytical results
- Help non-technical users understand data
- Provide actionable business insights

3.3 Data Visualization

The system automatically generates appropriate visualizations based on the user's question.

Possible visualizations include:

- Bar Charts
- Line Charts
- Pie/Donut Charts
- Area Charts
- Scatter Plots
- KPI Cards
- Heatmaps

For example:

- Product comparison → Bar Chart
- Monthly revenue → Line Chart
- Regional contribution → Pie/Donut Chart
- Sales relationship → Scatter Plot

3.4 Trend Analysis

The system identifies important trends within business data.

Output may include:

- Increasing sales trends
- Decreasing revenue trends
- Product growth patterns
- Regional performance trends
- Seasonal patterns

Purpose:

- Understand business performance over time
- Identify positive and negative trends
- Support strategic planning

3.5 Anomaly Detection

The system detects unusual or abnormal values within the dataset.

Examples:

- Unexpected sales spikes
- Sudden revenue drops
- Unusual transaction values
- Abnormal regional performance
- Unexpected changes in product demand

Purpose:

- Identify potential business problems
- Detect unusual transactions or performance
- Support further investigation

3.6 Comparative Analysis

The system compares business performance across different dimensions.

Examples:

- Product vs Product
- Region vs Region
- Month vs Month
- Quarter vs Quarter
- Current Year vs Previous Year
- Actual vs Target

Purpose:

- Identify high- and low-performing segments
- Understand performance differences
- Support business decisions

3.7 Actionable Recommendations

Based on identified patterns and KPIs, the system can provide possible business recommendations.

Examples:

- Focus on high-performing products
- Investigate regions with declining revenue
- Review products with consistently low sales
- Monitor unusual changes in demand
- Increase inventory for products showing strong growth

The recommendations are intended as **decision-support insights** rather than automatic business decisions.

3.8 Interactive Analytics Dashboard

The system provides a centralized dashboard containing:

- KPI Cards
- Charts and Graphs
- Trend Indicators
- Anomaly Alerts
- Business Insights
- Query Results
- Dataset Summary

4. Novelty of the Proposed System

The proposed system enhances traditional business analytics by combining **Natural Language Interaction, Automated Data Analysis, KPI Generation, Visualization, Trend Detection, Anomaly Detection, and AI-Based Insight Generation** into a single platform.

4.1 Natural Language-Based Data Analytics

Unlike traditional analytics tools that require users to manually create formulas, queries, or dashboards, the proposed system allows users to interact with business data using natural language.

Users can ask questions in a conversational manner and receive analytical results without requiring advanced technical knowledge.

4.2 Automated KPI Identification

The system identifies relevant business metrics from the uploaded dataset and calculates appropriate KPIs automatically.

This reduces the need for users to manually create formulas for common business metrics.

4.3 AI-Based Query Understanding

The system interprets the user's natural-language question and determines:

- User intent
- Relevant dataset columns
- Required metrics
- Required aggregation
- Appropriate analysis
- Suitable visualization

This enables users to interact with structured data in a conversational way.

4.4 Automatic Visualization Selection

Instead of requiring users to manually select chart types, the system determines an appropriate visualization based on the analytical question.

For example:

“Show monthly revenue.”

→ Time-series analysis → Line chart

“Compare revenue by region.”

→ Comparative analysis → Bar chart

4.5 Automated Trend and Anomaly Detection

The system analyzes business data to identify significant trends and unusual observations automatically. This helps businesses discover problems or opportunities that may not be immediately visible in raw datasets.

4.6 Explainable Business Insights

The system does not only display numerical results. It also explains the findings in natural language. This creates a bridge between **raw business data and actionable decision-making**.

4.7 Conversational Data Exploration

Users can ask multiple questions about the same uploaded dataset without repeatedly performing manual analysis.

For example:

User: Which product has the highest revenue?

Assistant: Product A.

User: What about its performance last quarter?

Assistant: Product A's revenue increased by 18% compared with the previous quarter.

This enables a conversational approach to business analytics.

5. Features

1. CSV and Excel Data Upload
2. Automated Data Preprocessing
3. Natural Language Data Querying
4. Automatic KPI Calculation
5. AI-Based Query Understanding
6. Automated Visualization Generation
7. Trend Analysis
8. Anomaly Detection
9. Comparative Business Analysis
10. Business Performance Dashboard
11. Natural Language Insight Generation
12. Actionable Decision-Support Recommendations
13. Conversational Data Exploration

6. Methodology

Step 1: Dataset Upload

The user uploads a business dataset through the web application.

Supported formats include:

- CSV
- Excel

Step 2: Data Ingestion

The system reads the uploaded dataset and identifies:

- Number of rows and columns
- Column names
- Data types
- Numerical variables
- Categorical variables
- Date/time columns
- Missing values

Step 3: Data Preprocessing

The uploaded data is cleaned and prepared for analysis.

Processing may include:

- Handling missing values
- Removing duplicate records
- Data type conversion
- Date/time processing
- Numerical data validation
- Column standardization
- Outlier identification

Step 4: Dataset Understanding

The system analyzes the structure and meaning of the dataset.

It identifies relationships between columns such as:

- Revenue and Product
- Sales and Region
- Profit and Category
- Sales and Time
- Quantity and Product

This information is used to determine which fields are relevant for future queries.

Step 5: Natural Language Query Processing

The user submits a question in natural language.

For example:

“Which product generated the highest revenue?”

The system processes the query to identify:

- Intent → Ranking
- Metric → Revenue
- Dimension → Product
- Operation → Aggregation + Maximum

Step 6: Analytical Query Generation

The system converts the user's question into an appropriate data-analysis operation.

Depending on the query, the system may perform:

- Filtering
- Grouping
- Aggregation
- Sorting
- Statistical analysis
- Comparison
- Time-series analysis

Step 7: KPI & Statistical Analysis

The system calculates relevant KPIs and analytical metrics.

Examples include:

- Sum
- Average
- Minimum
- Maximum
- Percentage change
- Growth rate
- Profit margin
- Sales contribution

Step 8: Trend & Anomaly Analysis

The system analyzes historical data to identify:

- Trends
- Seasonal patterns
- Sudden changes
- Outliers
- Performance deviations

Step 9: Visualization Generation

Based on the user's query and analysis results, the system automatically generates an appropriate visualization.

For example:

Revenue over time → Line Chart

Revenue by product → Bar Chart

Regional contribution → Pie/Donut Chart

Step 10: Insight Generation

The analytical results are converted into natural-language business insights.

The generated response may contain:

- Key finding
- Supporting metric
- Trend explanation
- Possible contributing factors
- Suggested action

Step 11: Dashboard Display

The system displays:

- KPIs
- Visualizations
- Analytical results
- Trends
- Anomalies
- AI-generated insights

through an interactive dashboard.

Step 12: Conversational Follow-Up

Users can continue asking questions about the uploaded dataset.

The system maintains the analytical context to support follow-up questions and deeper exploration.

7. Models & Technologies Used

Natural Language Query Understanding

1. Large Language Model (LLM)

- Understands natural-language business questions
- Identifies user intent
- Determines relevant dataset fields
- Converts questions into analytical operations
- Generates natural-language explanations

2. Sentence Transformers / Embedding Model

- Converts queries and dataset metadata into vector representations
- Supports semantic matching between user questions and dataset columns
- Helps identify relevant fields for analytical queries

Data Processing

1. Pandas

- CSV and Excel processing

- Data cleaning
- Filtering
- Grouping
- Aggregation
- Statistical analysis

2. NumPy

- Numerical computation
- Mathematical operations
- Statistical processing

Visualization

1. Plotly / Matplotlib

- Interactive charts
- Business dashboards
- Trend visualization
- Comparative visualization

Trend & Statistical Analysis

1. Statistical Analysis

- Mean and median analysis
- Percentage change
- Growth-rate calculation
- Correlation analysis
- Distribution analysis

2. Time-Series Models

Depending on the dataset and project scope:

- ARIMA
- Exponential Smoothing

These can be used for time-series forecasting and future business-performance analysis.

Anomaly Detection

1. Isolation Forest

- Detects unusual observations
- Identifies abnormal sales or revenue values
- Supports business anomaly analysis

2. Statistical Outlier Detection

- Identifies observations that significantly deviate from expected patterns
- Supports detection of unusual business performance

Application / Dashboard

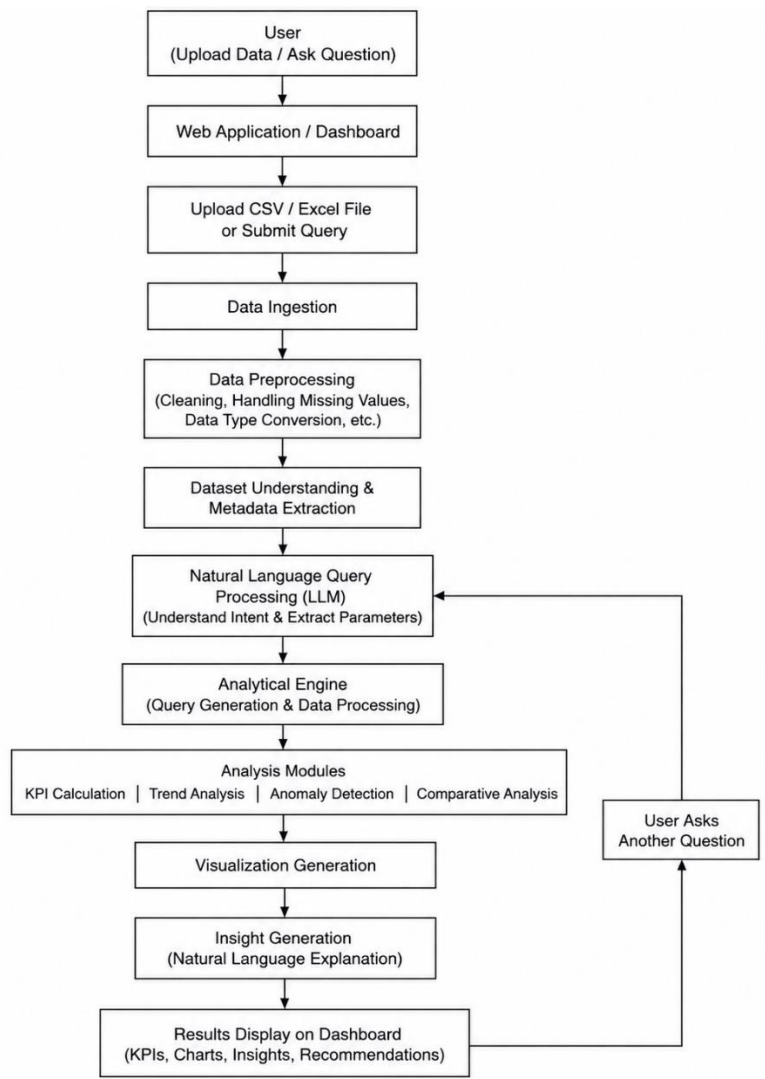
Possible technologies include:

- Python
- Streamlit / Flask / FastAPI
- HTML, CSS, JavaScript
- SQL Database
- REST APIs

The exact framework can be selected according to the final implementation requirements.

8. System Architecture

The proposed system follows the following architecture:



9. Team Member Responsibilities

Team Member	Responsibility
Vanshika	Data Preprocessing, Feature Engineering, AI/ML Analysis, KPI Logic, Model Integration, Testing and Evaluation
Muskan	Dataset Collection, Data Cleaning, Visualization Design, Dashboard Components and Documentation
Harshit	Backend Development, API Integration, Database Management and System Deployment
Anshul	Natural Language Query Processing, UI Development and Visualization Integration

Collaborative Responsibilities

All team members will jointly contribute to:

- Testing and validation
- System integration
- Model evaluation
- Dashboard testing
- Performance optimization

- Documentation
- Final presentation
- Project demonstration

10. Expected Outcomes

The proposed **Business KPI Assistant** is expected to:

- Simplify business data analysis.
- Allow users to query datasets using natural language.
- Automatically calculate important business KPIs.
- Reduce the need for manual formulas and data analysis.
- Generate relevant business visualizations automatically.
- Identify important trends and patterns.
- Detect unusual or abnormal business performance.
- Provide understandable AI-generated business insights.
- Support comparative analysis across products, regions, and time periods.
- Help users identify potential business problems and opportunities.
- Provide decision-support information through an interactive dashboard.
- Improve the speed and accessibility of data-driven decision-making.

11. Conclusion

The proposed **Business KPI Assistant** integrates Natural Language Processing, Large Language Models, Data Analytics, Statistical Analysis, Visualization, KPI computation, Trend Analysis, and Anomaly Detection to simplify traditional business analytics.

By allowing users to upload Excel or CSV datasets and ask questions using natural language, the system reduces the technical barrier associated with conventional data analysis. Instead of requiring users to manually write formulas, perform calculations, or create charts, the assistant automatically analyzes the data, generates relevant KPIs and visualizations, identifies trends and anomalies, and explains the results in an understandable manner.

The system therefore provides an intelligent and interactive approach to business analytics, helping organizations transform raw business data into meaningful insights and faster, data-driven decision-making.